Week 5: Login and your first feature

Plan your data on paper, then build it in Softr



ENT207TC Digital Startup Lab | Bogdan Marculescu and Stefan Seedorf

GROUND RULES FOR TOOLS

Don't pay for any of these apps

1. **The free plans are enough for everything in this module.**
2. Paying won't improve your grade. We assess what your team **understands and can show** us.
3. Skip the AI builder features, like Softr's AI Co-Builder and Vibe Coding block.
   Build with the visual editor.
4. If a tool asks you to upgrade, click NO; **do not pay anything**.

AI as a feature inside your product, using the API keys we give you, is welcome. AI building the product for you in one prompt is not.

Today's plan: three hours

0:00

Data model basics

20 min

Talk and examples

0:20

Paper data model

30 min

Draw it, then swap and check

0:50

Map your app

20 min

Pages and buttons, on paper

1:10

Tables in Softr

40 min

Build what is on your paper

1:50

Login

20 min

Turn it on and test it

2:10

Start Feature 1

50 min

Plan it, start it, update Kanban and log

By the end of this week: a working login and Feature 1 in your Softr app

Three rules for today



Build the app yourself

Create the tables, pages and login by hand. Do not use Generate with AI to make the app for you.



Softr AI is for the AI step only

Use Softr AI credits **only** when you build the AI feature, and keep one account's free credits for Demo Day.



You must be able to explain what you built

AI help is allowed, but using AI without understanding the result counts as academic dishonesty (Module Handbook, section 17).

Split your screen: chat AI and Softr

Your chat AI

My Bookings page shows the Machine name but not its Room. What should I check?

Check that Machine is a Linked Record field. Then add a Lookup field that brings in Room.

Why a Lookup and not a copy?

Ask a question...

Softr

Interface | **Data** | Users | Workflows

Machine	Room	Status
Washer 1	B12	Free
Washer 2	B12	Booked
Dryer 1	C03	Free

Lookup: Room

You add this field yourself

Left: ask why, ask what an error means, ask for another way.

Right: you click, type and build every table and field yourself.

AI as your lab partner

Use it to

Explain a concept or a field type in plain words.

Explain an error message.

Suggest another way when you are stuck.

Check your logic and your data model.

Ask better questions

Weak: Fix my app.

Better: My Bookings page shows nothing for a new user. The table has records and I am logged in. What should I check first?

The AI will not do the engineering for you. Asking clear, specific questions is a skill you are practicing.

PART 1

Plan your data on paper

Before you open Softr: what does your app need to remember?

An app is a set of linked spreadsheets

1 **Table**

A list of one kind of thing. Example: Machines.

2 **Field**

One column. One detail about every item, such as Room.

3 **Record**

One row. One single item, such as Washer 1.

1 **Table: Machines**

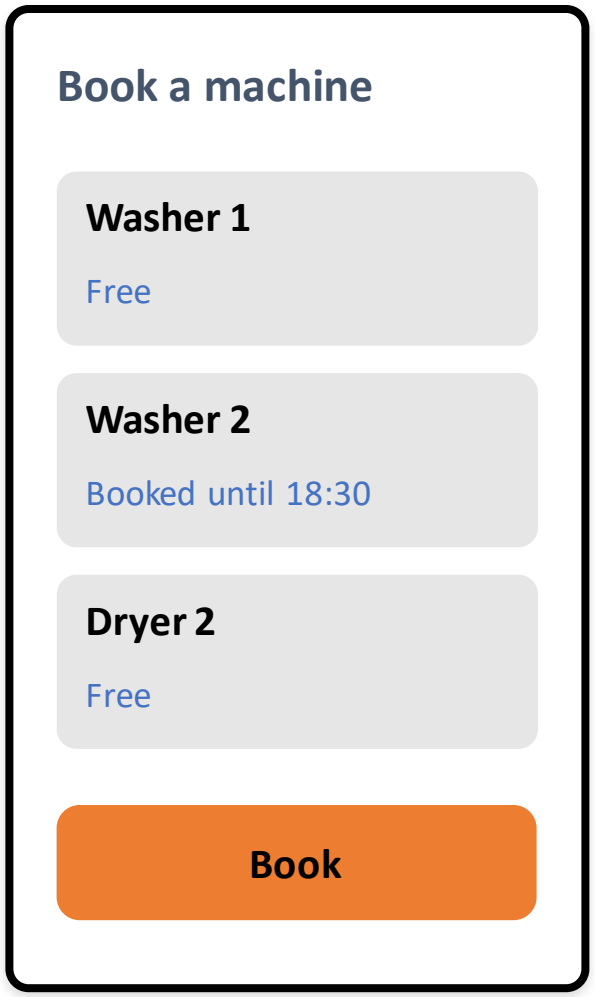
2

3

Name	Room	Type	Working?
Washer 1	Block A	Washer	Yes
Washer 2	Block A	Washer	Yes
Dryer 1	Block B	Dryer	No
Dryer 2	Block B	Dryer	Yes

In Softr you will see exactly this: tables to choose from, fields as columns, records as rows.

What the user sees and what is stored



reads from

Machines

Name	Status
Washer 1	Free
Washer 2	Booked

Bookings

Machine	Until
Washer 2	18:30
Dryer 1	20:00

Every screen reads from a table.

Every button adds to or changes a table.

Step 1: list the things your app keeps track of

EXAMPLE:



Users

One row per person who uses the app



Machines

One row per washer or dryer



Bookings

One row per booking



How to find your tables

Write your problem in one sentence and underline the nouns. If you would keep a separate list for it, it is a table. Aim for 3 to 5 tables this semester.

Your app's spreadsheets

Before you open Softr, write down the lists your app needs to keep. Example: a study-room booking app:

Students

Name	Major	Year
Li Wei	Business	2
Mei Lin	Finance	3

Bookings

Student	Room	Date	Time
Li Wei	A101	12 Oct	14:00
Mei Lin	B204	13 Oct	10:00

Each booking points to one student and one room.

Rooms

Room	Building	Seats
A101	Library	6
B204	CB	10

1 What lists does your app need to keep?

2 Give each list 3–5 columns.

3 Which lists point to each other?

Put it in your Weekly Log before you build. You build these lists yourself in Softr's free plan; no AI credits needed.

Step 2: give each table its fields

Table: Bookings

Date	Time slot	Status	Note	User	Machine
Date	*Select*	*Select*	*Long Text*	*Linked Record*	*Linked Record*
14 Nov	18:00-19:00	Booked	Own powder	Mia Chen	Washer 2
15 Nov	10:00-11:00	Done		Leo Park	Dryer 1

Select

A short, fixed list of answers, like Status

Date

Real dates, not text, so you can sort and filter

Number

Anything you will add up or compare

Linked Record

Points to a record in another table

Step 3: link the tables

Users

Name	Email
Mia Chen	mia@uni.edu
Leo Park	leo@uni.edu

1 many

Bookings

Date	Status	User	Machine
14 Nov	Booked	Mia Chen	Washer 2
15 Nov	Done	Leo Park	Dryer 1

many 1

Machines

Name	Room
Washer 2	Block A
Dryer 1	Block B

Link it

Bookings has two linked fields: User and Machine. Mia lives in one place (Users) and every booking points to her.

Do not copy it

Typing 'Mia Chen' into every booking row. One typo and you have two people. Change her email once and the old rows are wrong.

Step 4: test it with three real situations

1

Mia books Washer 2 for 18:00.

A new record goes into Bookings. Fill Date, Time slot and Status. Link User to Mia and Machine to Washer 2.

2

Mia cancels her booking.

Which field changes? Status becomes Cancelled. The record stays, so you keep the history.

3

The manager wants to see every booking for Washer 2.

Open the Washer 2 record in Machines and read its linked bookings.

If you cannot answer from your paper, a table, field or link is missing. Fix the paper now. It is far cheaper than fixing Softr later.

Paper task: draw your data model

20 min

1. Name your 3 to 5 tables. Sticky notes first if you like.
2. Under each table, list its fields and write the field type next to each.
3. Draw an arrow for every link and mark it Linked Record.
4. On the back, write 3 test situations and walk each through the sheet.
5. Take a photo of the sheet. It goes on your Kanban card and in your weekly log.

Pencil, one sheet per group. Every member writes at least one table, so your own contribution shows.



Check with another group

10 min

Group A's sheet



Group B's sheet

Write your feedback in three lines on their sheet. Then take your sheet back, read it, and fix your model.

Mentors and TAs: ask one group, 'Show me where a booking is stored.'

Three questions for the sheet

1. Can you find where each of their three situations is stored?
2. Is any information typed twice?
3. Which record should you link?

Paper task 2: map your app

20 min

Example: the laundry booking app

Page	What it shows (table: fields)	Button → what happens	Who can open it
Machines	Machines: Name, Room, Status	Book → opens the Book form	Any logged-in user
Book form	Machines: Name. Bookings: Date, Time slot	Save → adds a record to Bookings with Status Booked	Any logged-in user
My bookings	Bookings: Date, Time slot, Status (only your own)	Cancel → sets Status to Cancelled	Any logged-in user
All bookings	Bookings: all fields, with User and Machine	Mark done → sets Status to Done	Admin only

1. One row for every page in your sketches
2. For every button: what it does and which table changes
3. For every list or form: the table and fields it uses
4. Who can open it: guest, user or admin

Check: every field on a page is in your data model. Every button changes a table or opens a page.
Take a photo of your map and keep it as proof of your work.

PART 2

Build it in Softr

Same tables, same fields, same links. Now on screen. ONE MASTER ACCOUNT (leader)

Rule: Softr AI is for the AI step only. Build the app itself by hand.

Your Softr trial lasts 7 days

A new Softr account starts with a 7-day Business trial.

Features marked Trial work for 7 days, then stop. If you sign up in Week 5, they are gone in Week 6.

1. Build only with features that work on the Free plan.
2. If an option shows a Trial badge, do not use it.
3. Find your end date: open your workspace, then Plan & billing.

How paid options look in Softr

Softr Databases	
Airtable	
Microsoft Excel	TRIAL (BUSINESS)
Xano	TRIAL (PRO)
Monday.com	TRIAL (PRO)
Supabase	TRIAL (PRO)

Example from the Add data source screen. Yours may differ.

Spend AI credits on purpose

Build accounts

Use AI credits only to build and test your AI feature.

Run it on one record first and read the credit cost before you run it on many.

Never use credits to make tables, pages or login. Build those by hand.

One demo account per group

Pick one account as your demo account.

Its free AI credits are only for Demo Day.

Do not run any AI in it before then.

Where to look: the AI credits card in Plan & billing shows what is left and has an Activity history link.

From your paper to Softr

On your paper		In Softr
A table	→	A table in Softr Databases
A field and its type	→	A field: Short Text, Number, Date, Select and so on
A record (a row)	→	A record
An arrow between two tables	→	A Linked Record field
A screen	→	A page, built from blocks
Who may log in	→	Users menu, then the Authentication tab

Softr Databases is built into Softr Studio.

Step 1: create your account and a new app

5 min

1. Sign up at softr.io on the free plan. Every student makes an account.
2. The group leader creates the app and invites teammates so everyone can edit.
3. Choose to start from scratch. **Do not choose Generate with AI.**
4. Name the app after your project.

Button names change from time to time. Follow what you see on screen, and keep the 'from scratch' option.



Upload Proof

Softr dashboard showing a new app from scratch

Step 2: create your tables

15 min

1. Open Softr Databases.
2. Create a database for your app.
3. Add one table for each table on your paper. Use the same names.
4. Add each field and choose its type from your sheet.

Keep the paper next to the laptop. Tick each field off as you add it.



Upload Proof

Softr Databases: your tables listed, fields as columns

Pick the right field type

Short Text

Names and titles. Up to 1,024 characters.

Long Text

Descriptions and notes. Up to 65,536 characters.

Number

Whole numbers or decimals.

Date

A date, or a date and time, with a picker.

Select

Pick one or more from a color-coded list.

Checkbox

Yes or no.

File

Images, PDFs, documents. 128 MB per file.

Linked Record

Points to records in another table.

Later, not today: Formula calculates a value. Lookup shows a value from a linked record. Rollup counts or sums linked records.

Step 3: link your tables

10 min

1. In the table that holds the link (for example Bookings), add a field.
2. Choose Linked Record as the type.
3. Pick the table it points to (for example Machines).
4. Open a record in the other table and check the link shows up there too.

Softr's own example links Clients to Projects and Projects back to Clients. Both directions work.



Upload Proof

A Linked Record field being set up: choosing the table it points to

Step 4: add at least five test records

10 min

1. Add five or more records that look real. No 'asdf' rows.
2. Make at least two bookings that link to different users and different machines.
3. Fill every field type at least once.
4. Check each link opens the right record.

Real-looking data makes testing Feature 1 much easier, and it looks better on Demo Day.

Upload Proof

A table with five or more realistic records

Step 5: turn on login

20 min

1. Open the Users menu and go to the Authentication tab.
2. Turn Login on.
3. Enable Email. Users can sign in with a password, a one-time code, or both.
4. Under Sign Up, pick Open so your 10+ testers can join later.
5. Softr creates the sign-up and log-in pages for you.

Skip SSO (it is a paid feature). Skip Magic Links: anyone holding the link can log in as that user.



Upload Proof

Users menu, Authentication tab: Login on, Email on, Sign Up setting

Step 6: build a page that shows your data

This week

1. Open the Pages menu.
2. You build application pages. Softr makes login pages for you.
3. Add a page, for example Machines.
4. Add a block that lists records and connect it to your table.
5. Choose which fields to show, then add the page to the navigation.

Block names change as Softr updates. Pick the block that shows records from a table, and check the data source.



Upload Proof

Pages menu with a new page and a block connected to a table

Step 7: Feature 1, add or change data

This week

Add record

Creates a new record, such as a new booking.

Edit record

Changes the details of a selected record.

One-click update

Applies a set change in one click, such as Booked to Done.

Delete record

Removes a record.

Set on List, Grid, Kanban, Table, Calendar, Inbox or Item details blocks.



Upload Proof

Block settings: the Add record or Edit record action

What counts as a feature

Counts

- A tracker with progress logic, such as a booking that moves from Booked to Done
- Search, filter or sort that users can use on a table
- Different views for admin and user
- A profile that changes what the user sees

Does not count

- Login on its own
- A static page or landing page
- A generic contact form
- A calendar with no logic behind it
- Good looks with no behavior

Write one sentence first: 'A user can _____ so that _____.'

Step 8: preview, publish and test

This week

User A: Mia

Her booking: Washer 2, 14 Nov

User B: Leo

His booking: Dryer 1, 15 Nov

Log in as each person. Does each one see what they should, and nothing more?

1. Use Preview, then Publish (top right of Studio).
2. Sign up with a second email address.
3. Try bad input: an empty form, a wrong date.
4. Ask a classmate to use the app with no help. Write down where they get stuck.

Upload Proof

Preview

Ring the bell to celebrate your successes!

Show your work

- Login works: sign up and log in with a second account
- Feature 1 works live, not just in a screenshot
- Photos of your paper data model and app map, and Softr tables that match it
- Kanban board updated, with at least one card moved to Done
- Weekly log (Form 1) uploaded with each person's reflection
- AI use written down: what you asked and what you changed

Everyone teaches one part

Each person owns some parts

Split the app between you: data model, Feature 1, app map, testing. Each of you becomes the expert in some part and teaches it to the group.

Prepare to explain

Be ready to show your part and say why it works, not only what to click. Take notes on both. A mentor may ask any group member.

Document your journey

Write down what confused you, which errors you hit and how you fixed them. These notes should go into your weekly log.

After this week: how was it? Remember to reflect on what happens.

Hand in this week, and what comes next

This week

- Working login
- Feature 1 working
- Photos of your paper data model and app map
- Kanban board updated
- Weekly log with individual reflections

Next week (Week 6)

- Integration and a second feature
- Target: a working app with 2 features
- Plan your AI feature